Eduvos
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Eduvos Assessment Coversheet

Academic Details

Campus: **Eduvos Online**

Faculty: **Information Technology**

Module Code: **ITBDA4-14**

Lecturer's Name: **Gilbert Lekula**

Group (if applicable) _____

Assessment Details

Assessment Type: **Project**

Assignment/Brief Number: **2**

Due Date/ Submission Date: **13 November 2026**

6Student Details (Group, if applicable)

#	Full Name	Student Number	Contact Number	% Participation	Signature
1					
2					
3					
4					
5					
6					

Student Details (Individual, if applicable)

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Student Number: **EDUV6693976**

Academic Integrity

AI Disclosure Completed [X] **Yes** [ ] **No**

Declaration:

I declare that this assessment is my own original work except for source material explicitly acknowledged. I confirm that this work has not previously been submitted for assessments in any course. I acknowledge the institutional AI policy and confirm that AI tools have only been used as stipulated and documented in the AI Disclosure Appendix.

Student Signature 

Date **08 November 2026**

Lecturer Use Only

Lecturer Comments:

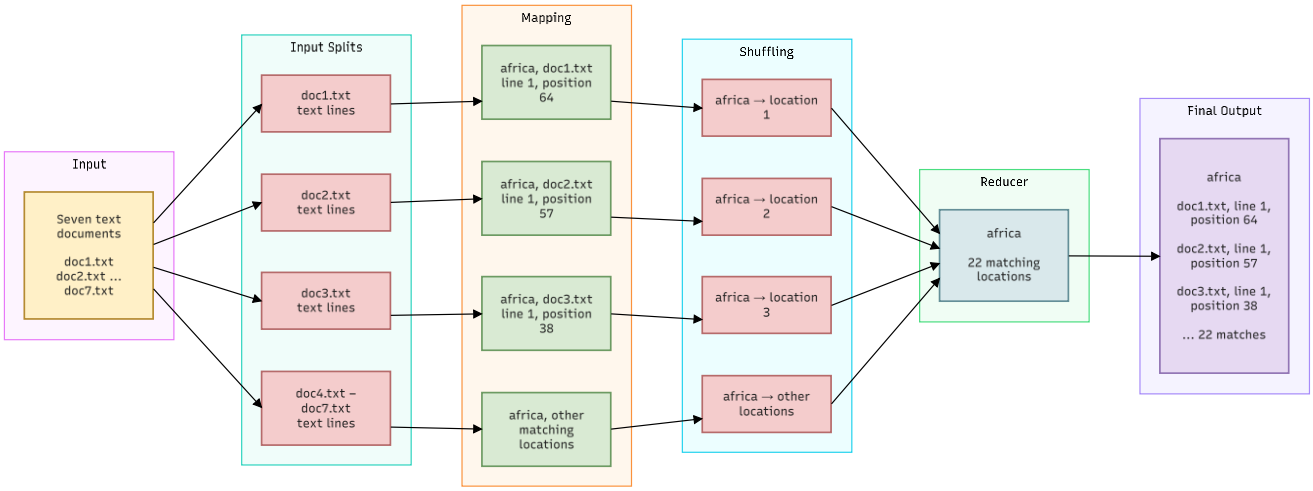
Marks Awarded (%):

Lecturer Signature:

Date:

Question 1

1.1



The MapReduce framework is used to search for a keyword specified by users, in this case across 7 documents, and to return the exact location of all keyword matches.

Input: In this case, it starts with the seven documents provided as input, where a user enters a keyword which is stored in the Hadoop job configuration.

Input splitting: Hadoop then divides the split into smaller ones, where each split gets assigned a Mapper for independent processing.

Mapping: Each Mapper reads every line of the text and searches for the keyword. The Mapper then converts the text and keyword to lowercase to ensure a case-insensitive search. Every match generates a key-value pair whereby the keyword is the key, and the value is the document name, line number and character position.

Shuffling and sorting: Hadoop then groups all the intermediate values with the same keyword under one key and makes the grouped result available to the Reducer.

Reducing: The Reducer gets the Keyword and the matching locations and then writes each location to the output without doing any additional calculations

Final Output: Finally, the Reducer produces the final output, which contains the searched keyword along with its corresponding locations in the input documents, allowing the user to locate where the keyword occurs across the documents.

1.2

```
import java.io.IOException;
import org.apache.hadoop.io.LongWritable;
import org.apache.hadoop.io.Text;
```

```
import org.apache.hadoop.mapreduce.Mapper;
import org.apache.hadoop.mapreduce.lib.input.FileSplit;

public class KeywordMapper extends Mapper<LongWritable, Text, Text, Text> {
    private String keyword;
    private String documentName;
    private int lineNumber = 0;
    @Override
    protected void setup(Context context) {
        keyword = context.getConfiguration().get("search.keyword").toLowerCase();
        documentName = ((FileSplit) context.getInputSplit()).getPath().getName();
    }

    @Override
    protected void map(LongWritable key, Text value, Context context)
            throws IOException, InterruptedException {
        lineNumber++;
        String line = value.toString();
        String lowerLine = line.toLowerCase();
        int position = 0;
        while ((position = lowerLine.indexOf(keyword, position)) != -1) {
            String location = documentName + ", line " + lineNumber
                    + ", character position " + (position + 1);
            context.write(new Text(keyword), new Text(location));
            position++;
        }
    }
}
```

1.3

```
import java.io.IOException;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Reducer;
public class KeywordReducer extends Reducer<Text, Text, Text, Text> {
    @Override
    protected void reduce(Text key, Iterable<Text> values, Context context)
            throws IOException, InterruptedException {
        for (Text value : values) {
            context.write(key, value);
        }
    }
}
```

1.4

```
import java.util.Scanner;
import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Job;
import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
```

```
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;

public class KeywordSearch {
    public static void main(String[] args) throws Exception {
        if (args.length != 2) {
            System.out.println("Usage: KeywordSearch <input folder> <output folder>");
            System.exit(1);
        }

        Scanner input = new Scanner(System.in);
        System.out.print("Enter keyword to search for: ");
        String keyword = input.nextLine().trim();

        if (keyword.isEmpty()) {
            System.out.println("Please enter a keyword.");
            input.close();
            System.exit(1);
        }
        Configuration configuration = new Configuration();

configuration.set("search.keyword", keyword);
configuration.set("mapreduce.framework.name", "local");
configuration.set("fs.defaultFS", "file:///");
configuration.set("hadoop.tmp.dir", "C:/hadoopsetup/tmp");
configuration.set("mapreduce.jobtracker.staging.root.dir",
        "C:/hadoopsetup/tmp/staging");
        Job job = Job.getInstance(configuration, "Keyword Search");
        job.setJarByClass(KeywordSearch.class);
        job.setMapperClass(KeywordMapper.class);
        job.setReducerClass(KeywordReducer.class);
        job.setOutputKeyClass(Text.class);
        job.setOutputValueClass(Text.class);

        FileInputFormat.addInputPath(job, new Path(args[0]));
        FileOutputFormat.setOutputPath(job, new Path(args[1]));

        input.close();
        System.exit(job.waitForCompletion(true) ? 0 : 1);
    }
}
```

```
Enter keyword to search for: africa
 africa  doc1.txt, line 3, character position 95
 africa  doc1.txt, line 3, character position 11
 africa  doc1.txt, line 1, character position 64
 africa  doc3.txt, line 54, character position 44
 africa  doc3.txt, line 52, character position 86
 africa  doc3.txt, line 50, character position 142
 africa  doc3.txt, line 48, character position 112
 africa  doc3.txt, line 46, character position 7
 africa  doc3.txt, line 40, character position 146
 africa  doc3.txt, line 5, character position 7
 africa  doc3.txt, line 3, character position 7
 africa  doc3.txt, line 1, character position 38
 africa  doc5.txt, line 48, character position 217
 africa  doc5.txt, line 38, character position 27
 africa  doc5.txt, line 23, character position 121
 africa  doc5.txt, line 3, character position 88
 africa  doc5.txt, line 1, character position 62
 africa  doc2.txt, line 58, character position 81
 africa  doc2.txt, line 58, character position 5
 africa  doc2.txt, line 33, character position 24
 africa  doc2.txt, line 3, character position 32
 africa  doc2.txt, line 1, character position 57
```

Question 2

2.1

```
library(stringr)
library(stopwords)

news_files <- list.files(path = "sa_news",pattern = "^news_[0-9]+\\.txt$",full.names = TRUE)
length(news_files)


#2.1

filetext_prep <- function(file_name) {

  text <- tolower(paste(readLines(file_name, warn = FALSE), collapse = " "))
  text <- gsub("[^a-zA-Z\\s]", " ", text)
  text <- gsub("\\s+", " ", text)
  words <- unlist(strsplit(trimws(text), "\\s+"))
  words <- words[!words %in% stopwords::stopwords("en")]

  paste(words, collapse = " ")
}

news_df <- data.frame(col_1 = basename(news_files),col_2 = sapply(news_files, filetext_prep),
  stringsAsFactors = FALSE)

head(news_df)
```

data.frame(file = news_df$col_1,words = sapply(strsplit(news_df$col_2, "\\s+"), length),characters = nchar(news_df$col_2))

```
> head(news_df)
                        col_1
sa_news/news_1.txt   news_1.txt
sa_news/news_10.txt news_10.txt
sa_news/news_2.txt   news_2.txt
sa_news/news_3.txt   news_3.txt
sa_news/news_4.txt   news_4.txt
sa_news/news_5.txt   news_5.txt

... <truncated>
sa_news/news_1.txt
eskom former official service provider appear c ... <truncated>
sa_news/news_10.txt
wet cold weather expected continue western cape monday july cold wet weather expected continue western cape week series cold fronts still make landfall coming days series cold fronts still track make landfall resulting week wet cold
conditions monday see coldest daytime temperatures next cold front rain expected arrive tuesday mm rain expected western parts mm can expected southern western mountains western cape local government environmental affairs developmen
t planning department said department added provincial disaster management centre pdmc monitoring situation south african weather service confirmed outlook strong gale force west north westerly winds south west coast emphasised high
danger runaway wildfires garden route district department water sanitation continues monitor river dam levels expected rise significantly coming week eskom reported several outages past hours caused severe weather cape town west coa
st overberg regions hardest hit f ... <truncated>
sa_news/news_2.txt
gauteng education commits implementing mayibuye primary school remedial actions tuesday july gauteng department education gde committed implementing remedial actions public protector report construction mayibuye primary school repor
t result investigation allegations functionaries gauteng department infrastructure development gdid well province s education department built sc ... <truncated>
sa_news/news_3.txt  new labour minister charts way forward tuesday july newly appointed employment labour minister nomakhosazana meth says department facing mammoth task ensuring continues earnest usher hope citizens substantive pos
itive change lives meth speaking ministerial committee meeting welcome new principals present department s programme together deputy ministers phumzile mgcina jomo sibiya meth held meeting executive committee including department s
acting director general onke mjo department s deputy director generals monday s meeting also attended heads compensation fund cf unemployment insurance fund uif well heads entities supported employment enterprises see commission con
ciliation mediation arbitration ccma productivity sa national economic labour council nedlac address meth said south africa still struggling stubborn triple challenges poverty unemployment inequality said scourge unemployment remain
s huge hurdle country facing official unemployment rate currently standing expanded unemployment rate according quarterly labour force survey qlfs quarter picture becomes dire comes youth unemployment rate youth aged years standing
youth aged years standing statistics remind us stakes still high must hasten strengthen implementation interventions dent unemployment rate meth maintained substantive results can achieved intensive collaborative mechanism propel us
realise department s objectives priorities said carrying forth mandate department guided set six priorities seventh administration priority put south africa work priority build industries inclusive economy priority tackle prohibitiv
e cost living priority invest people priority defend democracy advance freedom priority better africa world priorities must find expression work department employment labour family priority create sustain million work opportunities
delivering public goods services communities implement cross cutting industrial strategy drives growth creates opportunity youth unemployed peo ... <truncated>
sa_news/news_4.txt
... <truncated>
sa_news/news_5.txt
president ramaphosa sends condolences ... <truncated>
> data.frame(file = news_df$col_1,words = sapply(strsplit(news_df$col_2, "\\s+"), length),characters = nchar(news_df$col_2))
          file words characters
1   news_1.txt   143       1098
2  news_10.txt   274       2014
3   news_2.txt   172       1428
4   news_3.txt   377       3031
5   news_4.txt   135        973
6   news_5.txt   146       1088
7   news_6.txt   255       1985
8   news_7.txt   158       1347
9   news_8.txt   444       3699
10  news_9.txt   220       1730
>
```

2.2

```
#creating the document-term matrix
library(tm)
library(topicmodels)

corpus <- Corpus(VectorSource(news_df$col_2))
dtm <- DocumentTermMatrix(corpus)
inspect(dtm)

#applying LDA
lda_model <- LDA(dtm, k = 5, control = list(seed = 2026))

#top 15 words:
top_words <- terms(lda_model, 15)
top_words

#the words distributions
topic_probabilities <- as.data.frame(posterior(lda_model)$terms)

topic_dfs <- list()
for (i in 1:5) {

  topic_df <- data.frame(term = top_words[, i],probability = as.numeric(topic_probabilities[i,
top_words[, i]]))

  topic_df <- topic_df[order(topic_df$probability, decreasing = TRUE), ]
  topic_dfs[[paste("Topic", i)]] <- topic_df
}
topic_dfs
```

```
<<DocumentTermMatrix (documents: 10, terms: 1114)>>
Non-/sparse entries: 1474/9666
Sparsity           : 87%
Maximal term length: 19
Weighting          : term frequency (tf)
Sample             :
    Terms
Docs coalition days department government invoices national public said service south
  1          0    0          0          0        0        0      0    2       4     1
  10         0    0          2          0        0        0      5    8       2     4
  2          0    1          6          2        0        1      0    5       1     2
  3          0    0          5          0        0        0      3    2       0     0
  4          0    0         10          2        0        3      1    5       2     3
  5          0    0          0          0        0        0      0    1       1     1
  6          0    0          0          0        0        1      0    2       1     2
  7          0   11          0          2       17        7      2    6       7     0
  8          0    0          0          0        0        0      9    5       2     5
  9         13    1          1         14        0        5      6    8       1     0
```

```
> #top 15 words:
> top_words <- terms(lda_model, 15)
> top_words
      Topic 1          Topic 2      Topic 3          Topic 4          Topic 5
 [1,] "eskom"          "cape"       "government"     "said"           "invoices"
 [2,] "police"         "said"       "coalition"      "department"     "said"
 [3,] "report"         "cold"       "said"           "public"         "days"
 [4,] "school"         "weather"    "governments"    "africa"         "departments"
 [5,] "power"          "west"       "public"         "south"          "psc"
 [6,] "said"           "western"    "role"           "administration" "service"
 [7,] "service"        "expected"   "policy"         "unemployment"   "paid"
 [8,] "two"            "department" "relationships"  "meth"           "national"
 [9,] "department"     "sandf"      "national"       "priority"       "public"
[10,] "court"          "members"    "governance"     "labour"         "gxoyiya"
[11,] "employee"       "south"      "finlay"         "employment"     "payment"
[12,] "investigation"  "north"      "political"      "service"        "officials"
[13,] "july"           "week"       "trust"          "country"        "non"
[14,] "actions"        "president"  "including"      "fund"           "providers"
[15,] "gde"            "african"    "administration" "people"         "within"
```

$`Topic 1`

	term	probability
1	eskom	0.025171625
2	police	0.018306636
3	report	0.016018307
4	school	0.013729977
5	power	0.011441648
6	said	0.011441648
7	service	0.011441648
8	two	0.011441648
9	department	0.011441648
10	court	0.009153318
11	employee	0.009153318
12	investigation	0.009153318
13	july	0.009153318
14	actions	0.009153318
15	gde	0.009153318

$`Topic 2`

	term	probability
1	cape	0.01946472
2	said	0.01703163
3	cold	0.01703163
4	weather	0.01703163
5	west	0.01703163
6	western	0.01703163
7	expected	0.01459854
8	department	0.01459854
9	sandf	0.01459854
10	members	0.01216545
11	south	0.00973236
12	north	0.00973236
13	week	0.00973236
14	president	0.00973236
15	african	0.00729927

$`Topic 3`

	term	probability
1	government	0.032036613
2	coalition	0.029748284
3	said	0.018306636
4	governments	0.018306636
5	public	0.013729977
6	role	0.013729977
7	policy	0.013729977
8	relationships	0.013729977
9	national	0.011441648
10	governance	0.011441648
11	finlay	0.011441648
12	political	0.011441648
13	trust	0.011441648
14	including	0.009153318
15	administration	0.009153318

```
$`Topic 4`
             term probability
1            said 0.019230769
2      department 0.019230769
3          public 0.019230769
4          africa 0.017307692
5           south 0.015384615
6  administration 0.015384615
7    unemployment 0.015384615
8            meth 0.013461538
9        priority 0.013461538
10         labour 0.011538462
11     employment 0.009615385
12        service 0.007692308
13        country 0.007692308
14           fund 0.007692308
15         people 0.007692308

$`Topic 5`
          term probability
1     invoices  0.03711790
2         said  0.03056769
3         days  0.02401747
4  departments  0.02183406
5          psc  0.02183406
6      service  0.01965066
7         paid  0.01746725
8     national  0.01528384
9       public  0.01528384
10     gxoyiya  0.01528384
11     payment  0.01528384
12   officials  0.01528384
13         non  0.01310044
14   providers  0.01310044
15      within  0.01310044
```

2.3

```
library(wordcloud)
library(RColorBrewer)
library(ggplot2)

set.seed(2026)

for (i in 1:5) {

  topic <- topic_dfs[[i]]
  wordcloud( words = topic$term,freq = topic$probability,min.freq = 1,max.words =
15,random.order = FALSE,colors = brewer.pal(8, "Dark2"))
  title(paste("Word Cloud for Topic", i))
}

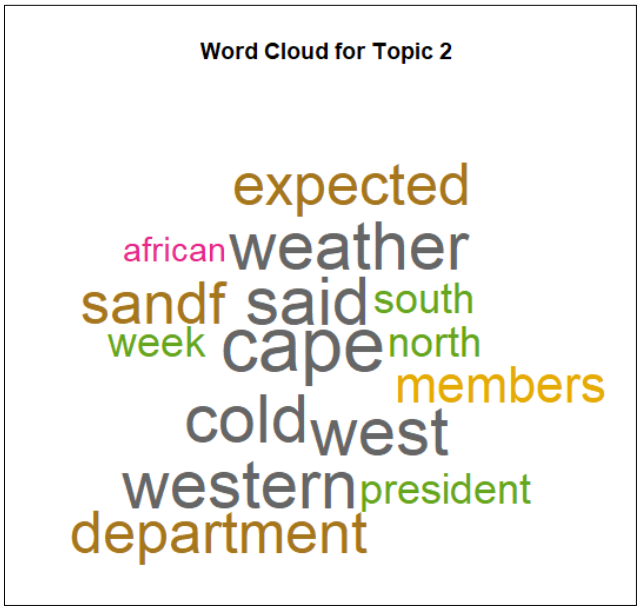
for (i in 1:5) {

  topic <- topic_dfs[[i]]

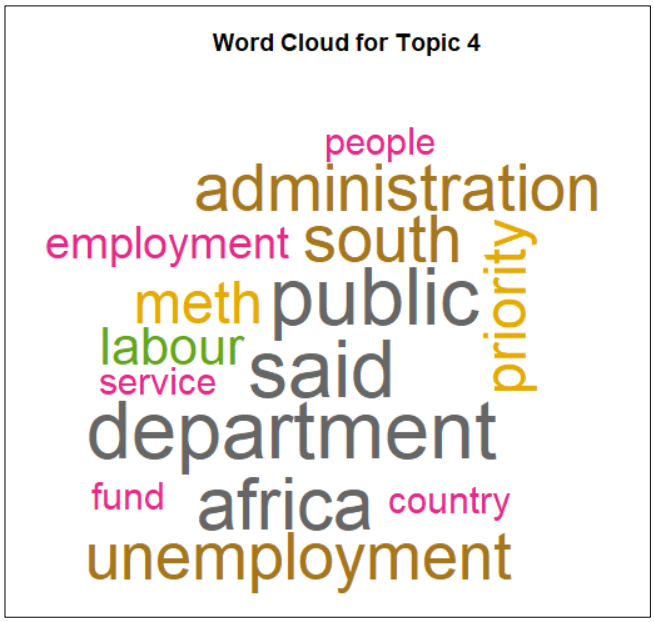
  p <- ggplot(topic, aes(x = reorder(term, probability), y = probability)) +
    geom_bar(stat = "identity", fill = "blue") +
    coord_flip() +
    labs(title = paste("Top Terms in Topic", i),x = "Term",y = "Probability") +theme_minimal()

  print(p) }
```

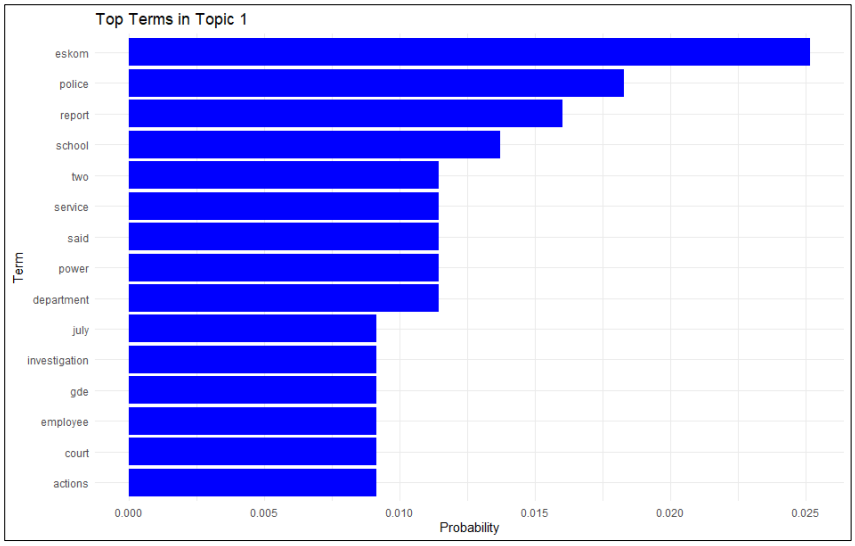

Word Cloud for Topic 1
actions
department
gde
power
service
two
school
eskom
court
july
said
police
report
investigation
employee

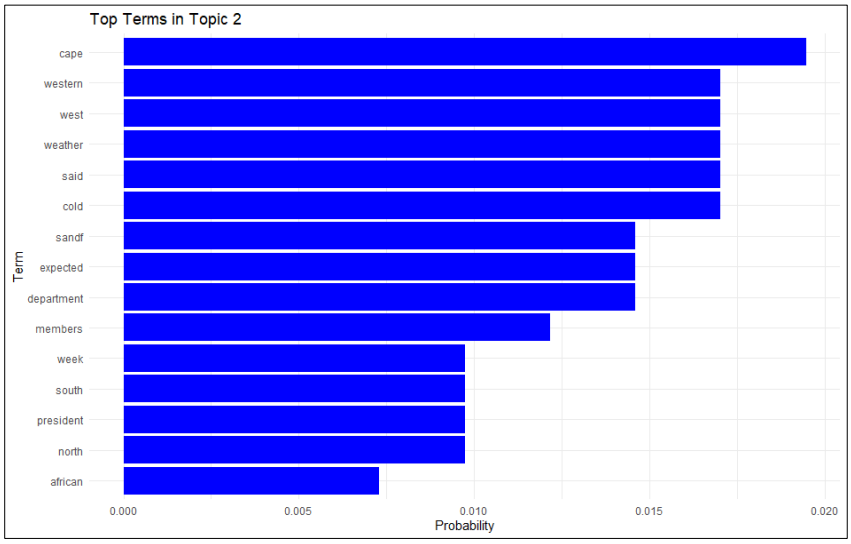

Word Cloud for Topic 2
expected
african
weather
sandf
said
south
week
cape
north
members
cold
west
western
president
department

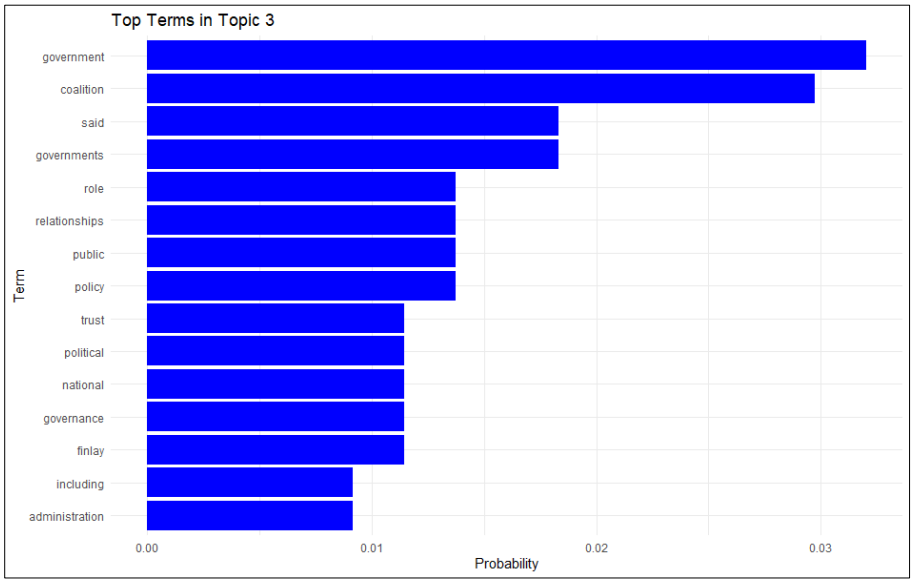

Word Cloud for Topic 3
administration
finlay
policy
role
including
national
coalition
government
trust
said
public
governments
political
relationships
governance

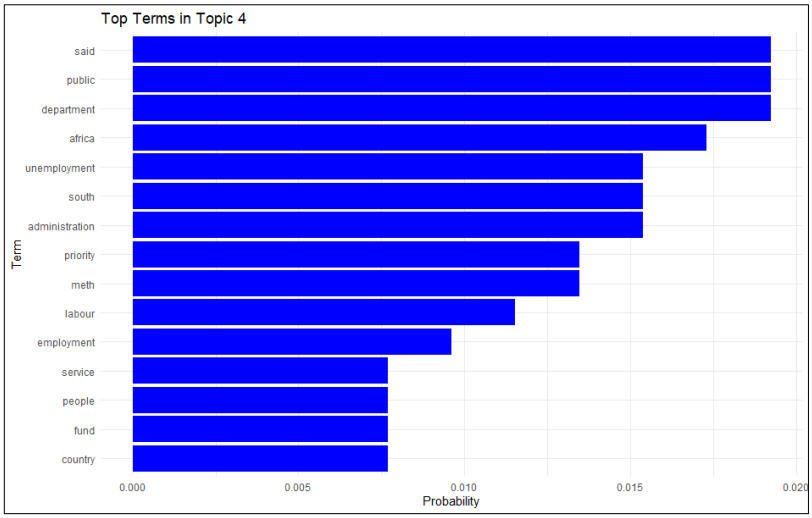

Word Cloud for Topic 4
people
administration
employment
south
meth
public
priority
labour
service
said
department
fund
africa
country
unemployment

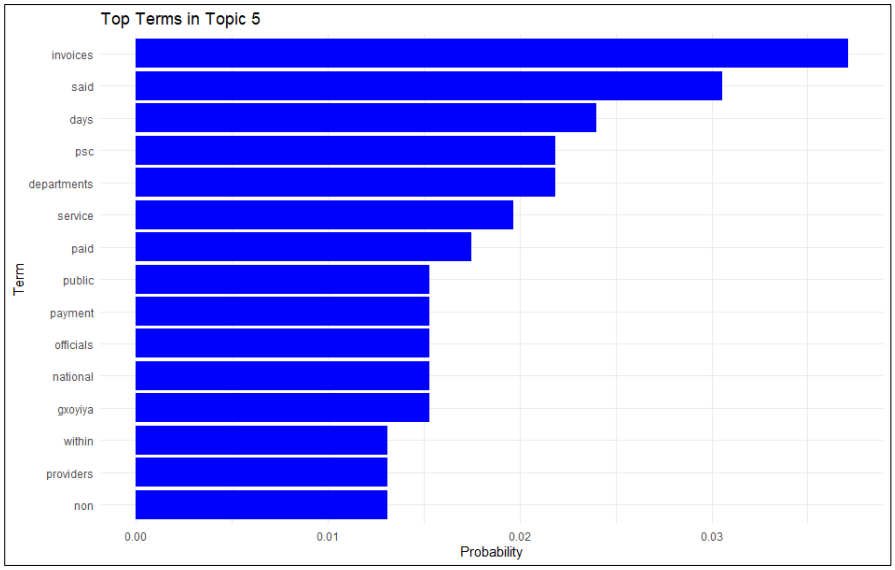

Word Cloud for Topic 5
officials
non
public
providers
service
paid
said
national
invoices
within
days
psc
departments
gxoyiya
payment


Top Terms in Topic 1
eskom
police
report
school
two
service
said
power
department
july
investigation
gde
employee
court
actions
Term
0.000
0.005
0.010
0.015
0.020
0.025
Probability


Top Terms in Topic 2
cape
western
west
weather
said
cold
sandf
expected
department
members
week
south
president
north
african
Term
0.000
0.005
0.010
0.015
0.020
Probability


Top Terms in Topic 3
government
coalition
said
governments
role
relationships
public
policy
trust
political
national
governance
finlay
including
administration
Term
0.00
0.01
0.02
0.03
Probability


Top Terms in Topic 4
said
public
department
africa
unemployment
south
administration
priority
meth
labour
employment
service
people
fund
country
Term
0.000
0.005
0.010
0.015
0.020
Probability


Top Terms in Topic 5
invoices
said
days
psc
departments
service
paid
public
payment
officials
national
gxoyiya
within
providers
non
Term
0.00
0.01
0.02
0.03
Probability

Question 3

3.1

```
url <- "https://groundup.org.za/article/a-decade-of-wasted-consultancies-how-slovo-park-exposes-south-africas-informal-settlement-upgrade-failure/"
webpage <- read_html(url)
# Extract article details
title <- webpage %>%
  html_element("h1") %>%
  html_text2()

author <- "Marie Huchzermeyer"
date <- "5 May 2026"
# Extract article paragraphs
paragraphs <- webpage %>%
  html_elements("article p") %>%
  html_text2()

# Remove empty paragraphs
paragraphs <- paragraphs[paragraphs != ""]

# Create an R data frame
article_df <- data.frame(title = title, author = author,date = date,paragraph = paragraphs, stringsAsFactors = FALSE)

head(article_df)

# Check the number of paragraphs
nrow(article_df)

# Check the extracted text
head(article_df$paragraph, 5)

# Check the total number of characters
sum(nchar(article_df$paragraph))
```

```
                                                
```

```
word_freq <- table(all_words)

#score each sentence using word frequencies
sentence_scores <- sapply(sentences, function(sentence) {

  sentence <- tolower(gsub("[^a-zA-Z\\s]", " ", sentence))
  words <- unlist(strsplit(sentence, "\\s+"))
  words <- words[words %in% names(word_freq)]

  if (length(words) == 0) {return(0)}

  mean(word_freq[words])
})

#select the five highest-scoring sentences
top_sentences <- data.frame(Sentence = sentences,Score = sentence_scores)
top_sentences <- top_sentences[order(top_sentences$Score, decreasing = TRUE), ]

head(top_sentences, 5)
```

```
                                                
```

```
# Calculate TF-IDF scores
tfidf_scores <- sapply(sentence_words, function(words) {

  if (length(words) == 0) {return(0)}

  tf <- table(words) / length(words)

  idf <- log(length(sentence_words) / (1 + sapply(names(tf), function(word) {
              sum(sapply(sentence_words, function(x) word %in% x))})))
  sum(as.numeric(tf) * idf)
})

# Create and rank the summary
tfidf_summary <- data.frame(Sentence = sentences,Score = tfidf_scores)

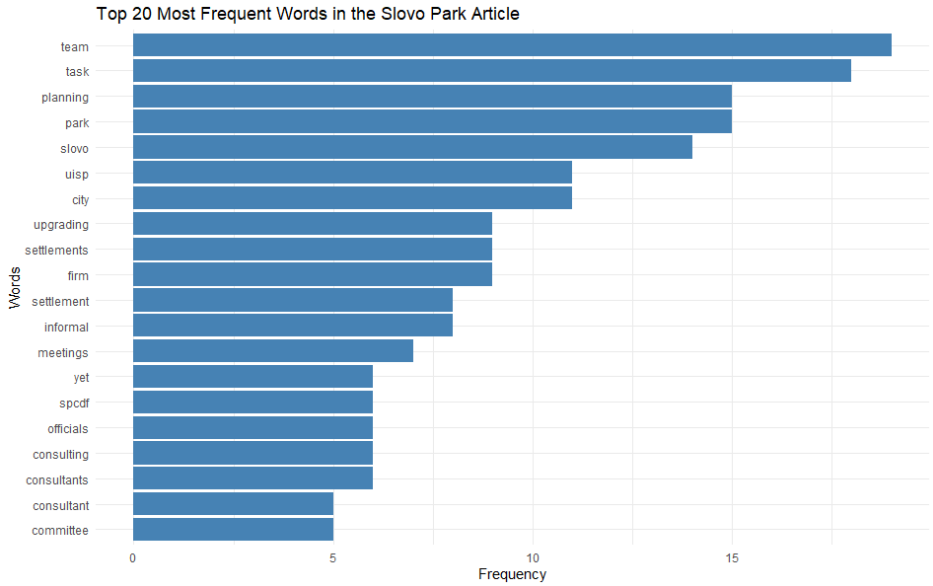
tfidf_summary <- tfidf_summary[order(tfidf_summary$Score, decreasing = TRUE),]

rownames(tfidf_summary) <- NULL

# Display the five most important sentences
head(tfidf_summary, 5)
```

```
                                                
```

labs(title = "Top 20 Most Frequent Words in the Slovo Park Article",x = "Words",y = "Frequency") +theme_minimal()



Generate a word cloud
library(wordcloud)
library(RColorBrewer)

wordcloud(words = names(word_freq),freq = as.numeric(word_freq),max.words = 30,
  colors = brewer.pal(8, "Dark2"),scale = c(3, 1),
  random.order = FALSE)



- The visualisations highlight the top themes in the article. The larger words such as team, task Planning, Slovo and task indicate higher frequency. The other important words include informal settlement, USIP, upgrading, consultants, and housing, which show the article's focus on informal settlement development and the challenges of upgrading Slovo Part.

Question 4

4.1

The main aim of this use case is to collect and analyse South African government news articles to identify numerical information reported in the news. The analysis focuses on extracting percentages from articles through regular expressions and examining the length of each article. The data source selected was SANews because it publishes articles related to public service and national developments. The scraped articles are unstructured textual data that can be processed and analysed using R.

```
# Load required libraries
library(rvest)
library(xml2)

# Read the two SAnews archive pages
page1 <- read_html("https://www.sanews.gov.za/south-africa")
page2 <- read_html("https://www.sanews.gov.za/south-africa?page=1")

# Extract article links from both pages
links1 <- page1 %>%
  html_elements("h2 a") %>%
  html_attr("href")

links2 <- page2 %>%
  html_elements("h2 a") %>%
  html_attr("href")

# Combine links and remove duplicates
article_urls <- unique (c(links1, links2))

# Convert links to complete URLs
article_urls <- url_absolute(article_urls, "https://www.sanews.gov.za")

# Create a data frame
article_data <- data.frame(article_url = article_urls,stringsAsFactors = FALSE)

# Extract the text from each article
get_text <- function(url) {
  page <- read_html(url)

  text <- page %>%
    html_elements("p") %>%
    html_text2() %>%
    paste (collapse = " ")

  return(text)
}

article_data$article_text <- sapply(article_data$article_url,get_text)
```

```
# Calculate the length of each article
article_data$article_length <- nchar(article_data$article_text)

# Use regular expressions to extract percentages
article_data$percentages <- lapply(
  article_data$article_text,
  function(x) {
    regmatches(x, gregexpr("[0-9]+%", x))[[1]]
  }
)

# Count the percentages found in each article
article_data$percentage_count <- lengths(article_data$percentages)

# Display the extracted data
head(article_data)
View(article_data)

# Check the number of articles collected
nrow(article_data)

# Check article text lengths
summary(article_data$article_length)

# Check the number of percentages extracted
summary(article_data$percentage_count)

# Check for missing and empty article text
sum(is.na(article_data$article_text))
sum(trimws(article_data$article_text) == "")

# Identify repeated spaces using regular expressions
spacing_issues <- sapply(article_data$article_text, function(x) {
  matches <- gregexpr(" {2,}", x)[[1]]
  if (matches [1] == -1) 0 else length(matches)
})

# Count articles containing repeated spaces
sum(spacing_issues > 0)

# Count repeated-space occurrences
sum(spacing_issues)

nrow(article_data)
length(unique(article_data$article_url))
```

	article_url	article_text	article_length	percentages	percentage_count
1	https://www.sanews.gov.za/south-africa/iec-achieves-91-completion-rate-local-election-results	The Electoral Commission (IEC) has achieved a 91% national result completion rate for the 2026 Lo...	2557	c("91%", "50%")	2
2	https://www.sanews.gov.za/south-africa/gcis-dg-visits-roc-observe-election-results-process	Government Communication and Information System (GCIS) Director-General Lloyd Nedohe has vi...	1951	character(0)	0
3	https://www.sanews.gov.za/south-africa/guidelines-strengthen-coalition-governance	Government is preparing guidelines for coalition negotiations and arrangements following the 202...	2452	character(0)	0
4	https://www.sanews.gov.za/south-africa/government-reiterates-warning-against-election-misinfor...	Government has reiterated its call for South Africans and political parties to refrain from spreading ...	3105	character(0)	0
5	https://www.sanews.gov.za/south-africa/shorter-queues-positive-voter-experiences-mark-2026-ele...	The 2026 Local Government Elections were characterised by shorter waiting times at voting station...	4583	c("95%", "91%", "90%", "90%", "91%", "95%", "79%", [...]	24
6	https://www.sanews.gov.za/south-africa/hsrc-survey-finds-voters-viewed-elections-free-and-fair	The overwhelming majority of voters regarded South Africa's 2026 Local Government Elections as f...	4325	c("90%", "91%", "95%", "95%", "83%", "81%", "85%", [...]	21
7	https://www.sanews.gov.za/south-africa/kzn-iec-captures-88-election-results-slips	The Electoral Commission of South Africa (IEC) in KwaZulu-Natal has captured 88% of the expected...	2411	c("88%", "82%", "64%", "40%", "70%", "63%", "64%", [...]	10
8	https://www.sanews.gov.za/south-africa/gauteng-disaster-teams-standby-severe-thunderstorms-lo...	The Gauteng Department of Cooperative Governance and Traditional Affairs (CoGTA) has activated ...	2200	character(0)	0
9	https://www.sanews.gov.za/south-africa/chikunga-pushes-accelerate-disability-employment	Government is stepping up efforts to expand employment opportunities for persons with disabilitie...	5220	c("7% ", "5% ", "56%")	3
10	https://www.sanews.gov.za/south-africa/mpumalanga-powering-south-africas-future-through-skills	By Moahlodi Maphori The Mpumalanga province has powered South Africa for generations. Today,...	3527	character(0)	0
11	https://www.sanews.gov.za/south-africa/trial-seven-accused-rapper-akas-murders-adjourned-april	The trial of the seven men charged with the murders of popular musician Kiernan 'AKA' Forbes and...	1037	character(0)	0
12	https://www.sanews.gov.za/south-africa/western-cape-reaffirms-commitment-agricultural-growth	The Western Cape government has reaffirmed its commitment to growing the agricultural sector th...	3779	character(0)	0
13	https://www.sanews.gov.za/south-africa/hot-and-humid-weather-expected-kzn	The South African Weather Service (SAWS) has warned that hot, humid weather will create extreme|...	1375	character(0)	0
14	https://www.sanews.gov.za/south-africa/20-nabbed-limpopo-election-related-offences	The South African Police Service (SAPS) in Limpopo has arrested 20 suspects for various election-re...	2406	character(0)	0
15	https://www.sanews.gov.za/south-africa/ige2026-law-enforcement-alert-results-capturing-continues	South Africa's law enforcement agencies remain on alert as the Electoral Commission of South Afric...	2667	character(0)	0
16	https://www.sanews.gov.za/south-africa/n-west-legislature-urges-voters-collect-ids-voter-registrati...	The North West Provincial Legislature Deputy Speaker, Linah Miga, has encouraged voters across t...	1958	character(0)	0
17	https://www.sanews.gov.za/south-africa/g20-development-group-meetings-kick-under-sas-leader...	South Africa has reiterated its intention of using its G20 Presidency to foster global solidarity, equal...	2540	character(0)	0
18	https://www.sanews.gov.za/south-africa/transnational-crimes-under-spotlight-regional-police-mee...	Law enforcement agencies within the Southern Africa region have committed to strengthening coll...	3696	character(0)	0
19	https://www.sanews.gov.za/south-africa/banyana-star-nominated-caf-award	Pretoria - Banyana Banyana and their Russian-based striker Nompumelelo Nyandeni have both bee...	1541	character(0)	0
20	https://www.sanews.gov.za/south-africa/mec-slams-report-tambo-memorial-hospital	Pretoria - Gauteng MEC for Health and Social Development, Qedani Mahlangu, said the statement ...	1304	character(0)	0
21	https://www.sanews.gov.za/south-africa/preparations-track-gauteng-matric-exams	Johannesburg - The Gauteng Department of Education says preparations to hold the 2009 matric e...	1998	character(0)	0
22	https://www.sanews.gov.za/south-africa/ige2026-results-capturing-making-steady-progress	Electoral Commission of South Africa (IEC) Chairperson Mosotho Moepya says results capturing an...	3122	c("60%", "87%", "69%", "64%", "60%", "52%", "37%", [...]	10
23	https://www.sanews.gov.za/south-africa/parties-urged-exercise-patience-kzn-election-results-are-c...	The Electoral Commission of South Africa (IEC) in KwaZulu-Natal has called on political parties, inde...	3131	c("35%", "3% ", "1% ", "49%")	4
24	https://www.sanews.gov.za/south-africa/election-observer-satisfied-voting-process	Defend Our Democracy (DOD) election observer Thandeka Sithole has expressed satisfaction with t...	2848	character(0)	0
25	https://www.sanews.gov.za/south-africa/iec-reports-159-million-election-participants-results-proce...	The Electoral Commission of South Africa (IEC) has reported that 15.9 million South Africans partici...	5176	character(0)	0
26	https://www.sanews.gov.za/south-africa/sa-welcomes-un-decision-advance-global-response-envir...	South Africa has welcomed a decision to strengthen international law enforcement efforts against f...	4635	37%	1
27	https://www.sanews.gov.za/south-africa/ige-2026-vote-counting-underway	Following a successful voting process on Wednesday, vote counting in the 2026 Local Government ...	3953	c("4%", "7%")	2
28	https://www.sanews.gov.za/south-africa/kzn-records-487-voter-turnout-results-capturing-continues	KwaZulu-Natal has recorded a voter turnout of 48.7% in the 2026 Local Government Elections, exce...	2890	c("7% ", "8% ", "50%", "18%", "12%")	5
29	https://www.sanews.gov.za/south-africa/revised-gdp-figures-improve-key-fiscal-ratios	National Treasury says Statistics South Africa's (Stats SA) 2026 rebasing and benchmarking of gross ...	2604	c("1% ", "6% ", "4% ", "70%", "6% ", "7% ", "8% ", [...]	10
30	https://www.sanews.gov.za/south-africa/parties-keep-close-watch-election-results-trickle	The Electoral Commission of South Africa's (IEC) national Results Operations Centre (ROC) in Midra...	895	character(0)	0
31	https://www.sanews.gov.za/south-africa/government-thanks-south-africans-participating-elections	Government has expressed its sincere appreciation to all South Africans, especially registered voter...	2707	character(0)	0
32	https://www.sanews.gov.za/south-africa/police-take-action-voting-day-linked-cases	Police have opened 57 dockets related to some 85 incidents that occurred on Voting Day of the Lo...	4796	character(0)	0
33	https://www.sanews.gov.za/south-africa/police-investigate-14-election-related-cases	The South African Police Service (SAPS) is investigating approximately 14 cases linked to the 2026 L...	3057	character(0)	0
34	https://www.sanews.gov.za/south-africa/voters-urged-use-remaining-time-cast-their-ballots	With just under two hours remaining before voting stations close at 9pm, the Electoral Commission...	4279	character(0)	0
35	https://www.sanews.gov.za/south-africa/ntuli-urges-citizens-vote-leaders-who-understand-their-n...	KwaZulu-Natal Premier Thamsanqa Ntuli has urged eligible voters to participate in the 2026 Local ...	2300	character(0)	0
36	https://www.sanews.gov.za/south-africa/young-voters-see-ballot-chance-shape-south-africas-future	Young South Africans have described voting as an important opportunity to influence the country's...	3935	character(0)	0
37	https://www.sanews.gov.za/south-africa/iec-enters-partnership-counter-disinformation	The Electoral Commission of South Africa (IEC) has entered into a partnership with Google, Meta, ...	5044	character(0)	0
38	https://www.sanews.gov.za/index.php/south-africa/parties-keep-close-watch-election-results-trickle	The Electoral Commission of South Africa's (IEC) national Results Operations Centre (ROC) in Midra...	895	character(0)	0
39	https://www.sanews.gov.za/index.php/south-africa/kzn-records-487-voter-turnout-results-capturin...	KwaZulu-Natal has recorded a voter turnout of 48.7% in the 2026 Local Government Elections, exce...	2890	c("7% ", "8% ", "50%", "18%", "12%")	5

```
> # Check the number of articles collected
> nrow(article_data)
[1] 41
>
> # Check article text lengths
> summary(article_data$article_length)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    895    2300    2848    2994    3935    5220
>
> # Check the number of percentages extracted
> summary(article_data$percentage_count)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  0.000   0.000   0.000   2.439   2.000  24.000
>
> # Check for missing and empty article text
> sum(is.na(article_data$article_text))
[1] 0
> sum(trimws(article_data$article_text) == "")
[1] 0
>
> # Identify repeated spaces using regular expressions
> spacing_issues <- sapply(article_data$article_text, function(x) {
+   matches <- gregexpr(" {2,}", x)[[1]]
+   if (matches[1] == -1) 0 else length(matches)
+ })
>
> # Count articles containing repeated spaces
> sum(spacing_issues > 0)
[1] 41
>
> # Count repeated-space occurrences
> sum(spacing_issues)
[1] 189
```

4.2

```
colSums(is.na(article_data))
#replacing repeated wwhitespace
article_data$article_text <- gsub("\\s+"," ",article_data$article_text)
article_data$article_text <- trimws(article_data$article_text)
```

```
#Recalculating article lengths.
spacing_issues <- sapply(article_data$article_text, function(x) {
  matches <- gregexpr(" {2,}", x)[[1]]
  if (matches[1] == -1) 0 else length(matches)
})

sum(spacing_issues)


article_data$article_length <- nchar(article_data$article_text)

#export cleaned csv
article_data$percentages <- sapply(article_data$percentages,paste,collapse = ", ")
write.csv(article_data, "sanews_cleaned.csv", row.names = FALSE)
```

```
[1] 189
> colSums(is.na(article_data))
     article_url     article_text   article_length      percentages percentage_count
               0                0                0                0                0
> colSums(is.na(article_data))
     article_url     article_text   article_length      percentages percentage_count
               0                0                0                0                0
> article_data$article_text <- gsub("\\s+"," ",article_data$article_text)
> article_data$article_text <- trimws(article_data$article_text)
> colSums(is.na(article_data))
     article_url     article_text   article_length      percentages percentage_count
               0                0                0                0                0
```

4.3

```
query2 = """
SELECT
    article_url,
    article_length,
    article_text
FROM sanews_articles
WHERE article_text REGEXP
'(?i)\\b(education|school|schools|university|universities|students|learners)\\b'
ORDER BY article_length DESC;
"""
pd.read_sql_query(query2, conn)
```

[11]:

	article_url	article_length	article_text
0	https://www.sanews.gov.za/south-africa/chikung...	5182	Government is stepping up efforts to expand em...
1	https://www.sanews.gov.za/south-africa/police-...	4783	Police have opened 57 dockets related to some ...
2	https://www.sanews.gov.za/south-africa/shorter...	4563	The 2026 Local Government Elections were chara...
3	https://www.sanews.gov.za/south-africa/mpumala...	3506	By Moahlodi Maphori The Mpumalanga province ha...
4	https://www.sanews.gov.za/south-africa/parties...	3117	The Electoral Commission of South Africa (IEC)...
5	https://www.sanews.gov.za/south-africa/prepara...	1986	Johannesburg - The Gauteng Department of Educa...

```
#What other topics are covered in the SAnews articles?

query2 = """
SELECT 'Economy and employment' AS topic, COUNT(*) AS total_articles
FROM sanews_articles
```

```
WHERE article_text REGEXP
'(?i)\\b(economy|economic|employment|unemployment|jobs|investment|inflation|GDP)\\b'

UNION ALL

SELECT 'Education', COUNT(*)
FROM sanews_articles
WHERE article_text REGEXP
'(?i)\\b(education|school|schools|university|universities|students|learners)\\b'

UNION ALL

SELECT 'Health', COUNT(*)
FROM sanews_articles
WHERE article_text REGEXP '(?i)\\b(health|hospital|hospitals|clinic|clinics|disease|vaccination)\\b'

UNION ALL

SELECT 'Agriculture', COUNT(*)
FROM sanews_articles
WHERE article_text REGEXP '(?i)\\b(agriculture|agricultural|farming|farmers|crops)\\b'

UNION ALL

SELECT 'Crime and safety', COUNT(*)
FROM sanews_articles
WHERE article_text REGEXP '(?i)\\b(crime|criminal|police|SAPS|arrest|arrests|safety)\\b'

UNION ALL

SELECT 'Weather and environment', COUNT(*)
FROM sanews_articles
WHERE article_text REGEXP '(?i)\\b(weather|rainfall|flood|flooding|climate|environment)\\b'

ORDER BY total_articles DESC;
"""

pd.read_sql_query(query2, conn)
```

	topic	total_articles
0	Crime and safety	11
1	Weather and environment	10
2	Economy and employment	7
3	Education	6
4	Health	2
5	Agriculture	1

The query returned six articles containing keywords related to education, such as schools, universities, students, learners, and the word "education" itself. The other set of results concerns the topic coverage in SABCNews articles, including crime and safety, weather and environment,

economy and employment, education, health, and agriculture. This demonstrates how regex can be used to retrieve articles on specific topics via SQL. However, it uses keywords in passing, so some topics may not be completely accurate.

4.3 The Link to GitHub: https://github.com/Xikongomelo/ITBDA4-14-Project-2

AI Disclosure Appendix

AI tool used	Purpose / how it was used	How I verified or adapted the output
ChatGPT 4o	Used to debug code if I could not understand the error and to find the root cause of the error.	Ran the code to see if it worked, but only the parts with errors; then rewrote the code related to the task.
Grammarly	Checking concordant agreement for the interpretations	Used it in moderation to still articulate what I was expressing.

Declaration:

I/we declare that this assessment is my/our own original work, except for source material explicitly acknowledged. I/we confirm that any use of Artificial Intelligence (AI) tools has been applied only as permitted by the assessment instructions and has been fully disclosed where required.

I/we understand that I/we remain responsible for the accuracy, originality, and academic quality of this submission. I/we am/are aware of the institution's policies on academic integrity and the requirements relating to the responsible use of AI in assessments.

I/we acknowledge that failure to comply with these requirements, including undeclared or inappropriate AI use, may result in academic penalties or disciplinary procedures in accordance with institutional policy.

Student Signature(s)	Date
	**8 November 2026**